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Rectangular secondary battery

Abstract

A rectangular battery with a plurality of stacked positive electrode plates. Each positive electrode plate includes a positive electrode active material arrangement portion and a positive electrode collector tab, the positive electrode active material arrangement portion including a plate-like core body and a positive electrode active material containing layer that is provided at the core body, the collector tab being integrated with the core body and protruding from the positive electrode active material arrangement portion. Each positive electrode collector tab includes a curved portion that is positioned between the corresponding positive electrode active material arrangement portion and the positive electrode collector terminal in a Z direction (a region in the Z direction indicated by B), and has at least a part thereof curved.

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Background/Summary

TECHNICAL FIELD
(1) The present disclosure relates to a rectangular secondary battery.
BACKGROUND ART
(2) FIG. **13** is a schematic sectional view of an upper side of a rectangular secondary battery **300** that is described in Patent Literature 1. As shown in FIG. **13**, the rectangular secondary battery **300** includes a rectangular outer casing **310**, an electrode body **311**, and a collector body **312**; and the electrode body **311** and the collector body **312** are disposed in the rectangular outer casing **310**. The electrode body **311** includes a plurality of stacked electrode plates **317**. Tabs **314** protrude from the corresponding electrode plates **317**, and the plurality of tabs **314** that protrude from the plurality of electrode plates **317** are disposed so as to overlap each other and constitute tab bundles **315**. Each tab bundle **315** is welded to the collector body **312** and is electrically joined to the collector body **312**. Tabs **314***a* that are flush with the electrode plates **317** from which they protrude exist in the tab bundles **315** in a region between the electrode body **311** and the collector body **312** (a region having a range indicated by G). Tension is applied to the tabs **314***a*, and the tabs **314***a* are in a state in which they are not loose at all.
(3) FIG. **14** is a schematic sectional view of an upper side of a rectangular secondary battery **400** that is described in Patent Literature 2. As shown in FIG. **14**, the rectangular secondary battery **400** includes a rectangular outer casing **410**, an electrode body (not shown), a lead **412**, and an auxiliary

lead **413**; and the electrode body, the lead **412**, and the auxiliary lead **413** are disposed in the rectangular outer casing **410**. The lead **412** and the auxiliary lead **413** constitute a collector body. In the rectangular secondary battery **400**, a tab bundle **415** that protrudes from the electrode body is interposed between the lead **412** and the auxiliary lead **413**, and is welded. As in the rectangular secondary battery **300**, in the rectangular secondary battery **400**, tabs **419** that are flush with electrode plates from which they protrude exist in the tab bundle **415** in a region between the electrode body and the auxiliary lead **413**. Tension is applied to the tabs **419**, and the tabs **419** are in a state in which they are not loose at all.

CITATION LIST

Patent Literature

(4) PTL 1: Japanese Published Unexamined Patent Application No. 2015-103318 PTL 2: Japanese Published Unexamined Patent Application No. 2015-130253

SUMMARY OF INVENTION

(5) In the rectangular secondary battery **300**, the tabs **314***a* that are subjected to tension and are not loose at all exist between the electrode body **311** and the collector body to which the tab bundles **315** are joined; and, in the rectangular secondary battery **400**, the tabs **419** that are subjected to tension and are not loose at all exist between the electrode body **311** and the collector body to which the tab bundle **415** is joined. Therefore, when the rectangular secondary batteries **300** and **400** are subjected to shock or vibration and a force acts upon the tab bundles **315** and the tab bundle **415** in an extension direction, the tabs **314***a* and the tabs **419** that are not loose at all are pulled and tend to be damaged.

(6) Accordingly, it is an object of the present disclosure to provide a rectangular secondary battery in which all tabs are unlikely to become damaged even if the rectangular secondary battery is subjected to shock or vibration.

(7) To this end, a rectangular secondary battery according to the present disclosure includes an electrode body that includes a plurality of stacked electrode plates; a rectangular outer casing that has an opening and that accommodates the electrode body; a cover plate that seals the opening and that is provided with an external terminal on a side opposite to a side of the electrode body; and a collector terminal that is electrically connected to the external terminal and that is disposed inside the rectangular outer casing, wherein each electrode plate includes an active material arrangement portion and a collector tab, the active material arrangement portion including a plate-like core body and an active material containing layer that is provided at the core body, the collector tab being integrated with the core body and protruding from the active material arrangement portion, wherein each collector tab includes a curved portion and an end, the curved portion being positioned between the active material arrangement portion corresponding thereto and the collector terminal in a height direction of the rectangular outer casing and having at least a part thereof curved, the end being positioned at the curved portion on a side opposite to a side of the active material arrangement portion, and wherein each end includes a joint portion that is joined and electrically connected to the collector terminal.

(8) In the present description, an external terminal is defined as a terminal portion of the rectangular secondary battery to which an external wire, constituted by, for example, a bus bar, is electrically connected. The height direction is defined as a direction normal to a plate-like portion of the cover plate including the plate-like portion; and a side where the external terminal exists is an upper side in the height direction, and a side opposite to the side of the external terminal is a lower side in the height direction.

(9) According to the rectangular secondary battery according to the present disclosure, all tabs can be made unlikely to become damaged even if the rectangular secondary battery is subjected to shock or vibration.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a perspective view of a rectangular secondary battery according to an embodiment of the present disclosure.
(2) FIG. **2** illustrates the rectangular secondary battery from a side thereof, and is a transparent view showing an internal structure.
(3) FIG. **3** is a partial sectional view showing an upper portion of a sectional view along line A-A of FIG. **2**.
(4) FIG. **4** is a plan view of a positive electrode plate of the rectangular secondary battery when seen in a thickness direction thereof.
(5) FIG. **5** is a plan view of a negative electrode plate of the rectangular secondary battery when seen in a thickness direction thereof.
(6) FIG. **6** is a perspective view of an electrode body of the rectangular secondary battery.
(7) FIG. **7** is a perspective view illustrating a structure of a part of a positive electrode collector terminal and a structure of a part of each positive-electrode-side cover (clamping portion) that are formed in an integral structure.
(8) FIG. **8** is a perspective view of a vicinity of a part of the positive electrode collector terminal in the rectangular secondary battery.
(9) FIG. **9** is an exploded perspective view of an electrical connection portion where the positive electrode collector terminal is electrically connected to a positive-electrode-side external terminal provided on an outer side of a cover plate.
(10) FIG. **10** is a perspective view illustrating the rectangular secondary battery where the cover plate to which, for example, the positive electrode collector terminal is fixed is mounted on a rectangular outer casing.
(11) FIG. **11** is a perspective view showing a part of an integral structure in which a collector terminal and each clamping portion are integrated with each other in a modification.
(12) FIG. **12** is a schematic view for illustrating an example of joining collector tabs to a collector terminal when the collector tabs are disposed so as to overlap each other to form one bundle.
(13) FIG. **13** is a sectional view illustrating a structure of a rectangular secondary battery of a first related art.
(14) FIG. **14** is a sectional view illustrating a structure of a rectangular secondary battery of a second related art.
DESCRIPTION OF EMBODIMENTS
(15) An embodiment according to the present disclosure is described in detail below with reference to the attached drawings. When, in the description below, for example, a plurality of embodiments and modifications are included, providing a new embodiment by combining features of the plurality of embodiments and modifications as appropriate is assumed from the outset. In the description and the drawings below, an X direction indicates a thickness direction of positive electrode plates **20** that are described below, and is the same as a stacking direction of the plurality of positive electrode plates **20**. A Y direction indicates one direction in a two dimensional plane in which the positive electrode plates **20** extend, and a Z direction indicates a height direction of a rectangular outer casing **11** that is described below and is the same as a direction that is normal to a plate-like portion of a cover plate **12** that is described below. The X direction, the Y direction, and the Z direction are orthogonal to each other.
(16) FIG. **1** is a perspective view of a rectangular secondary battery (hereunder simply referred to as “rectangular battery”) **1** according to an embodiment of the present disclosure. FIG. **2** illustrates the rectangular battery **1** from a side thereof, and is a transparent view showing an internal structure. FIG. **3** is a partial sectional view showing an upper portion of a sectional view along line

A-A of FIG. **2**. As shown in FIGS. **1** and **2**, the rectangular battery **1** includes the rectangular outer casing (rectangular outer can) **11**, the cover plate **12**, and a stack-type electrode body **14**. The rectangular outer casing **11** is made of, for example, a metal, desirably, aluminum or an aluminum alloy, and has an opening on an upper side in the Z direction. The rectangular battery **1** includes an insulating sheet (not shown), and the insulating sheet is disposed so as to cover an inner surface of the rectangular outer casing **11** excluding the opening side of the rectangular outer casing **11**. The rectangular outer casing **11** may be constituted by an insulator, such as synthetic resin, in which case, the insulating sheet can be omitted.

(17) Although described below, for example, the electrode body **14** is fixed to the cover plate **12**. After fixing, for example, the electrode body **14** to the cover plate **12**, the cover plate **12** is fitted to the opening of the rectangular outer casing **11**. By joining the cover plate **12** and a fitting portion of the rectangular outer casing **11** by, for example, laser welding, the cover plate **12** is integrated with the rectangular outer casing **11** to form a rectangular battery case **15**.

(18) As shown in FIG. **3**, the electrode body **14** includes the plurality of positive electrode plates **20**, a plurality of negative electrode plates **30**, and a plurality of separators (not shown). The number of positive electrode plates **20** and the number of negative electrode plates **30** are, for example, greater than or equal to 70 and less than or equal to 80. The negative electrode plates **30** are disposed on two sides of the electrode body **14**. The number of positive electrode plates and the number of negative electrode plates **30** may be less than **70** or greater than **80**. The electrode body **14** is accommodated inside the rectangular outer casing **11**. Next, the structures of the positive electrode plates **20**, the negative electrode plates **30**, and the electrode body **14** are described in detail. FIG. **4** is a plan view of a positive electrode plate **20** when seen in the thickness direction (the X direction) thereof. FIG. **5** is a plan view of a negative electrode plate **30** when seen in the thickness direction (the X direction) thereof. FIG. **6** is a perspective view of the electrode body **14**.

(19) As shown in FIG. **4**, each positive electrode plate **20** includes a positive electrode active material arrangement portion **21** and a positive electrode collector tab **22**. Each positive electrode active material arrangement portion **21** includes a plate-like positive electrode core body and a positive electrode active material containing layer **24** that is applied to two surfaces of the positive electrode core body. The positive electrode collector tabs **22** protrude from the positive electrode active material arrangement portions **21**. In the example shown in FIG. **4**, each positive electrode active material arrangement portion **21** has a substantially rectangular shape in side view. One pair of edges **25***a* of each positive electrode active material arrangement portion **21** extend substantially parallel to the Z direction, and the other pair of edges **25***b* thereof extend substantially in the Y direction. Each positive electrode collector tab **22** is made of a material that is the same as the material of each positive electrode core body and is integrated with its corresponding positive electrode core body. Each positive electrode collector tab **22** has a substantially rectangular shape in side view, and protrudes upward in the Z direction from one side in the Y direction of the upper edge **25***b* of its corresponding positive electrode active material arrangement portion **21**. One pair of edges of each positive electrode collector tab **22** extend substantially parallel to the Z direction, and the other pair of edges thereof extend substantially parallel to the Y direction.

(20) As shown in FIG. **5**, each negative electrode plate **30** includes a negative electrode active material arrangement portion **31** and a negative electrode collector tab **32**. Each negative electrode active material arrangement portion **31** includes a plate-like core body and a negative electrode active material containing layer **34** that is applied to two surfaces of the negative electrode core body. Each negative electrode collector tab **32** protrudes from its corresponding negative electrode active material arrangement portion **31**. In the example shown in FIG. **5**, each negative electrode active material arrangement portion **31** has a substantially rectangular shape in side view. One pair of edges **35***a* of each negative electrode active material arrangement portion **31** extend substantially parallel to the Z direction, and the other pair of edges **35***b* thereof extend substantially in the Y direction. Each negative electrode collector tab **32** is made of a material that is the same as the

material of each negative electrode core body and is integrated with its corresponding negative electrode core body. Each negative electrode collector tab **32** has a substantially rectangular shape in side view, and protrudes upward in the Z direction from the other side in the Y direction of the upper edge **35***b* of its corresponding negative electrode active material arrangement portion **31**. One pair of edges of each negative electrode collector tab **32** extend substantially parallel to the Z direction, and the other pair of edges thereof extend substantially parallel to the Y direction.
(21) Referring to FIG. **6**, the positive electrode plates **20** and the negative electrode plates **30** are alternately stacked via the separators with the positive electrode collector tabs **22** and the negative electrode collector tabs **32** being disposed so as to protrude upward in the Z direction. By the stacking, the electrode body **14** is formed. As shown in FIGS. **4** and **5**, the area of each negative electrode active material arrangement portion **31** in side view is larger than the area of each positive electrode active material arrangement portion **21** in side view. Although described below, the electrode body **14** is joined to a positive electrode terminal and a negative electrode terminal that are disposed inside the rectangular outer casing **11**, and is disposed at a predetermined position inside the rectangular outer casing **11**. When seen from the X direction with the electrode body **14** disposed at the predetermined position, all of the positive electrode active material arrangement portions **21** overlap the negative electrode active material arrangement portions **31**, and ring-shaped peripheral portions of the negative electrode active material arrangement portions **31** do not overlap the positive electrode active material arrangement portions **21**. The thickness of each positive electrode core body is larger than the thickness of each negative electrode core body.
(22) In a lithium-ion secondary battery, when the plate width of each positive electrode plate is larger than the plate width of each negative electrode plate, during charging, lithium ions that exit from a positive electrode active material of each positive electrode plate protruding from the plate width of each negative electrode plate concentrates on both end portions of each negative electrode plate, and a lithium dendrite tends to precipitate. The precipitation of lithium dendrite occurs more easily when the thickness of each negative electrode core body is larger than the thickness of each positive electrode core body. The precipitation of lithium dendrite causes deterioration of the battery, such as a reduction in the capacity maintaining rate when charging and discharging of electricity are repeated. In the present embodiment, by causing the thickness of each positive electrode core body to be larger than the thickness of each negative electrode core body and causing all of the positive electrode active material arrangement portions **21** to overlap the negative electrode active material arrangement portions **31**, precipitation of lithium dendrite is suppressed to suppress, for example, a reduction in the capacity maintaining rate when charging and discharging of electricity are repeated.
(23) The positive electrode core bodies and the positive electrode collector tabs **22** are made of, for example, an aluminum or an aluminum alloy foil. The positive electrode active material containing layers **24** can be formed by using, for example, lithium-nickel oxides as a positive electrode active material, acetylene black (AB) as a conductive agent, polyvinylidene fluoride (PVDF) as a binding agent, and N-methyl-2-pyrrolidone as a dispersion medium. When the positive electrode active material is described in more detail, as the positive electrode active material, any material may be selected as appropriate and used as long as the material is a compound that allows lithium ions to be reversibly occluded/discharged. As such positive electrode active materials, lithium transition metal composite oxides are desirable. For example, lithium transition metal composite oxides that are represented by LiMO.sub.2 (where M is at least one type selected from Co, Ni, and Mn) that allows lithium ions to be reversibly occluded/discharged, that is, LiCoO.sub.2, LiNiO.sub.2, LiNi.sub.yCo.sub.1-yO.sub.2 (y=0.01˜0.99), LiMnO.sub.2, or LiCoO.sub.xMn.sub.yNi.sub.zO.sub.2 (x+y+z=1); or LiMn.sub.2O.sub.4 or LiFePO.sub.4 may be used in such a way that one type is used singly or a plurality of types are mixed and used. Further, lithium cobalt composite oxides to which dissimilar metal elements such as zirconium, magnesium, aluminum, and tungsten are added are usable. However, the positive electrode active material

containing layers **24** may be made of any other publicly known materials other than those mentioned above.

(24) The positive electrode active material arrangement portions **21** are produced, for example, as follows. A positive electrode active material is mixed with, for example, a conductive agent and a binding agent, and this mixture is kneaded in a dispersion medium to produce a paste-like positive electrode active material slurry. Thereafter, the positive electrode active material slurry is applied to each positive electrode core body. Then, when the positive electrode active material slurry applied to the positive electrode core bodies is dried and compressed, the positive electrode active material arrangement portions **21** are formed.

(25) The negative electrode core bodies and the negative electrode collector tabs **32** are made of, for example, a copper or a copper alloy foil. The negative electrode active material of the negative electrode active material containing layers **34** is not particularly limited as long as the material is one that that allows lithium to be reversibly occluded/discharged. For example, carbon materials, silicon materials, lithium metal, metals that form an alloy with lithium or alloy materials thereof, or metal oxides can be used. From the viewpoint of material costs, it is desirable to use carbon materials as the negative electrode active material. For example, natural graphite, artificial graphite, mesophase pitch-based carbon fiber (MCF), mesocarbon microbeads (MCMB), coke, or hard carbon can be used. In particular, from the viewpoint of increasing high charge-discharge rate characteristics, as the negative electrode active material, it is desirable to use carbon materials formed by covering graphite material with low crystalline carbon.

(26) It is desirable to form the negative electrode active material containing layers **34** by using styrene-butadiene copolymer rubber particle dispersion (SBR) as a binding agent, carboxymethyl cellulose (CMC) as a thickening agent, and water as a dispersion medium. The negative electrode active material arrangement portions **31** are produced, for example, as follows. A negative electrode active material is mixed with, for example, a conductive agent and a binding agent, and this mixture is kneaded in a dispersion medium to produce a paste-like negative electrode active material slurry. Thereafter, the negative electrode active material slurry is applied to each negative electrode core body. Then, when the negative electrode active material slurry applied to the negative electrode core bodies is dried and compressed, the negative electrode active material arrangement portions **31** are formed.

(27) As the separators, publicly known separators that are generally used in nonaqueous electrolyte secondary batteries can be used. For example, it is desirable to use separators made of polyolefin. Specifically, not only separators made of polyethylene but also separators having a polypropylene layer formed on a polyethylene surface or separators having an aramid-based resin applied to a surface of polyethylene separators may be used.

(28) Inorganic filler layers may be formed at interfaces between the positive electrode plates **20** and the separators to interfaces between the negative electrodes **30** and the separators. As a filler, oxides or phosphate compounds using one or a plurality of, for example, titanium, aluminum, silicon, and magnesium; or those whose surface is treated with, for example, hydroxides can be used. The filler layers may be formed by directly applying slurry that contains the filler to the positive electrode plates **20**, the negative electrode plates **30**, or the separators; or may be formed by attaching sheets formed from the filler to the positive electrode plates **20**, the negative electrode plates **30**, or the separators.

(29) Referring again to FIG. **3**, the rectangular battery **1** further includes a positive electrode collector terminal **40**, metal positive-electrode-side covers **50**, serving as examples of clamping portions, and a resin insulating cover **60**. The positive electrode collector terminal **40** is disposed inside the rectangular outer casing **11** and extends in the X direction. The plurality of positive electrode collector tabs **22** that are included in the plurality of positive electrode plates **20** are separated in two and are disposed so as to overlap each other, and form two bundles. The two bundles each constitute a positive-electrode-side tab bundle **41**. The positive electrode collector

terminal **40** and each positive-electrode-side cover **50** are desirably made of aluminum, and are integrated with each other.

(30) FIG. **7** is a perspective view illustrating a structure of a part of the positive electrode collector terminal **40** and a structure of a part of each positive-electrode-side cover **50** and illustrating an integral structure thereof. As shown in FIG. **7**, the positive electrode collector terminal **40** includes a plate-like portion **45**. The rectangular battery **1** includes two positive-electrode-side covers **50**, and each positive-electrode-side cover **50** is connected to the positive electrode collector terminal **40** by a corresponding connecting portion **42**. The positive-electrode-side covers **50** and the connecting portions **42** have a rod-like shape and are integrated with each other, and extend in the X direction from one side end portion of the positive electrode collector terminal **40** in the Y direction. The thickness of each connecting portion **42** is smaller than the thickness of each positive-electrode-side cover **50**. One of the positive-electrode-side tab bundles **41** is caused to extend upward in the Z direction from a lower side of the positive electrode collector terminal **40** in the Z direction so as to contact one side edge portion in the X direction on one side of the positive electrode collector terminal **40** in the Y direction, and the other positive-electrode-side tab bundle **41** is caused to extend upward in the Z direction from the lower side of the positive electrode collector terminal **40** in the Z direction so as to contact the other side end portion in the X direction on one side of the positive electrode collector terminal **40** in the Y direction.

(31) Thereafter, with connection portions of the positive-electrode-side covers **50** that are connected with the corresponding connecting portions **42** being fulcra, the positive-electrode-side covers **50** are bent in XY planes so as to rotate in the directions of arrows C, and thereby a part of each positive-electrode-side tab bundle **41** is clamped by the corresponding positive-electrode-side cover **50** and the corresponding edge portion of the positive electrode collector terminal **40** in the X direction. Since the thickness of each connecting portion **42** is smaller than the thickness of each positive-electrode-side cover **50** and the connecting portions **42** are less rigid than the positive-electrode-side covers **50**, it is possible to smoothly bend the positive-electrode-side covers **50**.

(32) Thereafter, as shown in FIG. **8**, that is, a perspective view of a vicinity of a part of the positive electrode collector terminal **40** in the rectangular battery **1**, an end of each positive-electrode-side tab bundle **41** on a side opposite to the side of its corresponding positive electrode active material arrangement portion **21** turns back so as to extend in the X direction along an upper surface **46** of the plate-like portion **45** of the positive electrode collector terminal **40**. Although described in detail later, in the rectangular battery **1**, the plate-like portion **45** of the positive electrode collector terminal **40** is disposed substantially parallel to the plate-like portion of the cover plate **12**. A surface of the plate-like portion **45** on a side opposite to the side of the positive electrode active material arrangement portions **21** is a surface of the plate-like portion **45** on the side of the cover plate **12** in the Z direction, and constitutes the upper surface **46** of the plate-like portion **45**. After the turning back above, the positive electrode collector terminal **40**, the positive-electrode-side covers **50**, and the connecting portions **42** define recessed portions **43** that accommodate parts of the positive-electrode-side tab bundles **41**.

(33) As shown in FIG. **8**, each positive electrode collector tab **22** includes a flat portion **22***c* that extends along the upper surface **46** of the plate-like portion **45**. When seen from the Z direction, there exist overlapping portions **48** where all of the flat portions **22***c* of all of the positive electrode collector tabs **22** included in the positive-electrode-side tab bundles **41** overlap each other. The overlapping portions **48** are joined to the upper surface **46** of the plate-like portion **45** by, for example, ultrasonic welding, laser welding, TIG welding, or resistance welding, and are electrically connected to the plate-like portion **45**. In other words, a joint portion of each positive electrode collector tab **22** that is joined to the positive electrode collector terminal **40** is included in the overlapping portions **48**.

(34) It is desirable that the length of each overlapping portion **48** in a protruding direction of the positive electrode collector tabs **22** be greater than or equal to 3 mm because all of the positive

electrode collector tabs **22** that are included in the positive-electrode-side tab bundles **41** can be reliably joined to the plate-like portion **45**. However, the length may be less than 3 mm. It is desirable that the length of each overlapping portion **48** in the protruding direction of the positive electrode collector tabs **22** be less than or equal to 10 mm because material costs can be reduced and downsizing is easily realized. However, the length of each overlapping portion **48** in the protruding direction of the positive electrode collector tabs **22** may be greater than 10 mm.
(35) Referring to FIG. **2** again, the positive electrode collector terminal **40** includes an external terminal connecting portion **65** on a side opposite to the joint portion of each positive electrode collector tab **22** in the Y direction. The external terminal connecting portion **65** is electrically connected to a positive-electrode-side external terminal **18** provided on an outer side of the cover plate **12**. FIG. **9** is an exploded perspective view of an electrical connection portion thereof. As shown in FIG. **9**, the external terminal connecting portion **65** includes a circular columnar rivet portion **61** that protrudes upward from the plate-like portion **45**. The rivet **61** is passed through a through hole of a hollow disk-shaped insulator **62**, a through hole provided in the plate-like portion of the cover plate **12**, a through hole provided in an insulator **64**, and a through hole provided in the positive-electrode-side external terminal **18**. The rivet **61** is inserted with the insulating cover **60** (see FIG. **3**), made of an insulating material such as resin, disposed between the positive electrode collector terminal **40** and the cover plate **12**.
(36) After inserting the rivet **61**, the rivet **61** is crimped. When the rivet **61** has been crimped, the rivet **61**, the insulator **62**, the insulating cover **60**, the cover plate **12**, the insulator **64**, and the positive-electrode-side external terminal **18** are integrated with each other, and the rivet **61** of the positive electrode collector terminal **40** is electrically connected to the positive-electrode-side external terminal **18**.
(37) As described above, the positive electrode plates **20** are joined to the positive electrode collector terminal **40**. Therefore, by crimping the rivet **61**, the plurality of positive electrode plates **20**, the positive electrode collector terminal **40**, the insulating cover **60**, and the cover plate **12** are integrated with each other. Although not described, the plurality of negative electrode plates **30**, a copper negative electrode collector terminal (not shown), the insulating cover **60**, and the cover plate **12** are also integrated with each other by a structure similar to that on the positive electrode side. As a result, the electrode body **14**, the positive electrode collector terminal **40**, the negative electrode collector terminal, the insulating cover **60**, and the cover plate **12** are integrated into an integral structure. Although the case in which the positive electrode collector terminal **40** is integrated with the cover plate **12** by crimping is described, the positive electrode collector terminal may be fixed to the cover plate by other joining means such as welding.
(38) FIG. **10** is a perspective view illustrating the rectangular battery **1** when the integral structure **70** is mounted in the rectangular outer casing **11** whose inner surface is covered by the insulating sheet. As shown in FIG. **10**, with the electrode body **14** (see FIG. **6**) disposed on a lower side in the Z direction, the integral structure **70** is moved relative to the rectangular outer casing **11** in the Z direction indicated by arrow D, and a portion other than the cover plate **12** is inserted into the rectangular outer casing **11**. Thereafter, as described above, the cover plate **12** is joined to an opening-side edge portion of the rectangular outer casing **11** by, for example, laser welding.
(39) Referring to FIG. **3** again, the insulating cover **60** has a substantially U shape in cross section, and, when the integral structure **70** is mounted in the rectangular outer casing **11**, the insulating cover **60** is fitted to inner surfaces of a pair of side plates of the rectangular outer casing **11** that face each other in the X direction by press-fitting. With the cover plate **12** joined to the opening-side end portion of the rectangular outer casing **11**, parts of the positive-electrode-side tab bundles **41** are clamped by the corresponding positive-electrode-side covers **50** and the corresponding edge portions of the positive electrode collector terminal **40** in the X direction, and the positive-electrode-side covers **50** are subjected to a force on the side of the positive electrode collector terminal **40** in the X direction from the insulating cover **60**.

(40) With the cover plate **12** joined to the opening-side edge portion of the rectangular outer casing **11**, each positive electrode collector tab **22** includes a curved portion **22***a*, an extending portion **22***b* in the height direction, and the flat portion **22***c*. Each extending portion **22***b* in the height direction and each flat portion **22***c* constitute an end. Each curved portion **22***a* is positioned between its corresponding positive electrode active material arrangement portion **21** and the positive electrode collector terminal **40** in the Z direction (a region in the Z direction indicated by a range B), and has at least a part thereof curved. Each extending portion **22***b* in the height direction is connected to an end portion of its corresponding curved portion **22***a* on a side opposite to the side of the positive electrode active material arrangement portion **21**, and extends in the Z direction. Parts of the extending portions **22***b* in the height direction are clamped by the corresponding positive-electrode-side covers **50** and the corresponding edge portions of the positive electrode collector terminal **40** in the X direction. The flat portions **22***c* are connected to end portions of the corresponding extending portions **22***b* in the height direction on a side opposite to the side of the curved portions **22***a*, and extend along the upper surface **46** of the plate-like portion **45** of the positive electrode collector terminal **40**. The flat portions **22***c* include the joint portions that are joined to the positive electrode collector terminal **40**.
(41) When the mounting of the integral structure **70** into the rectangular outer casing **11** is completed, a nonaqueous electrolyte is injected via an injection hole (not shown) provided in the cover plate **12**. Thereafter, a predetermined charging operation is performed by using the positive-electrode-side external terminal **18** and a negative-electrode-side external terminal **19** (refer to FIG. **2**), and, after a reaction gas that is generated by a charging reaction of the battery has been previously generated, the electrolyte injection hole is hermetically sealed, so that the rectangular battery **1** is produced. The electrolyte injection hole is sealed by, for example, using a blind rivet or performing welding. The electrolyte injection hole is sealed in a dry air environment in which an inert gas atmosphere (rare gases such as N.sub.2 or Ar) or the moisture content is controlled. In this way, abnormal deterioration of the battery caused by moisture entering the battery case **15** and reacting with the electrolyte is prevented from occurring.
(42) Solvents of the nonaqueous electrolyte are not particularly limited, so that solvents that have been hitherto used in nonaqueous electrolyte secondary batteries can be used. For example, cyclic carbonates, such as ethylene carbonate (EC), propylene carbonate (PC), butylene carbonate, and vinylene carbonate (VC); chain carbonates, such as dimethyl carbonate (DMC), methyl ethyl carbonate (MEC), and diethyl carbonate (DEC); ester-containing compounds, such as methyl acetate, ethyl acetate, propyl acetate, methyl propionate, ethyl propionate, and γ-butyrolactone; sultone-group-containing compounds, such as propane sultone; ether-containing compounds, such as 1,2-dimethoxyethane, 1,2-diethoxyethane, tetrahydrofuran, 1,2-dioxane, 1,4-dioxane, and 2-methyltetrahydrofuran; nitrile-containing compounds, such as butyronitrile, valeronitrile, n-heptanenitrile, succinonitrile, glutarnitrile, adiponitrile, pimelonitrile, 1,2,3-propanetricarbonitrile, and 1,3,5-pentanetricarbonitrile; and amide-containing compounds, such as dimethylformamide, can be used. In particular, solvents in which a part of H of these substances is substituted by F are desirably used. The substances can be used singly, or a plurality of such substances can be used in combination. In particular, solvents in which a cyclic carbonate and a chain carbonate are combined with each other, or solvents in which a compound containing a small amount of nitrile or a compound containing ether is further combined thereto are desirably used.
(43) As nonaqueous solvents of nonaqueous electrolytes, ionic liquids can be used. In this case, cation types and anion types are not particularly limited. However, from the viewpoints of low viscosity, electrochemical stability, and hydrophobic property, the use in combination of, as cations, a pyridinium cation, an imidazolium cation, and a quaternary ammonium cation and, as an anion, a fluorine-containing imide-based anion is particularly desirable.
(44) Further, as solutes used in nonaqueous electrolytes, publicly known lithium salts that have hitherto been generally used in nonaqueous electrolyte secondary batteries can be used. As such

lithium salts, lithium salts containing one or more types of elements among P, B, F, O, S, N, and Cl can be used. Specifically, lithium salts, such as LiPF.sub.6, LiBF.sub.4, LiCF.sub.3SO.sub.3, LiN(FSO.sub.2).sub.2, LiN(CF.sub.3SO.sub.2).sub.2, LiN(C.sub.2F.sub.5SO.sub.2).sub.2, LiN(CF.sub.3SO.sub.2) (C.sub.4F.sub.9SO.sub.2), LiC(C.sub.2F.sub.5SO.sub.2).sub.3, LiAsF.sub.6, LiClO.sub.4, LiPF.sub.2O.sub.2, and mixtures of these salts can be used. In order to increase high charge-discharge rate characteristics and durability of nonaqueous electrolyte secondary batteries, in particular, it is desirable to use LiPF.sub.6.
(45) As solutes, a lithium salt in which an oxalate complex is used as the anion can also be used. As a lithium salt in which an oxalate complex is used as the anion, in addition to LiBOB (lithium-bis oxalate borate), there can be used a lithium salt having an anion in which C.sub.2O.sub.4.sup.2− is coordinated at a center atom, such as Li[M(C.sub.2O.sub.4).sub.xR.sub.y] (in the formula, M represents a transition metal or an element that is selected from the elements of Group XIII, Group XIV, and Group XV in the periodic table; R represents a group that is selected from halogen, an alkyl group, and a halogen-substituted alkyl group; x represents a positive integer; and y represents 0 or a positive integer). Specific examples include Li[B(C.sub.2O.sub.4)F.sub.2], Li[P(C.sub.2O.sub.4)F.sub.4], and Li[P(C.sub.2O.sub.4).sub.2F.sub.2]. However, in order to form a stable film on a negative electrode surface in a high-temperature environment, using LiBOB is most desirable.
(46) Not only may the solutes above be singly used, but also two or more types thereof may be mixed and used. Although the concentration of the solutes is not particularly limited, it is desirable that the concentration thereof be 0.8 to 1.7 moles per one liter of nonaqueous electrolyte. Further, in a usage requiring electric discharge for a large current, it is desirable that the concentration of the solutes be 1.0 to 1.6 moles per one liter of nonaqueous electrolyte.
(47) The rectangular battery **1** of the present disclosure includes the electrode body **14** that includes the plurality of stacked positive electrode plates **20**, the rectangular outer casing **11** that has an opening and that accommodates the electrode body **14**, the cover plate **12** that seals the opening and that is provided with the positive-electrode-side external terminal **18** on a side opposite to the side of the electrode body **14**, and the positive electrode collector terminal **40** that is electrically connected to the positive-electrode-side external terminal **18** and that is disposed inside the rectangular outer casing **11**. Each positive electrode plate **20** includes the corresponding positive electrode active material arrangement portion **21** and the corresponding positive electrode collector tab **22**, the positive electrode active material arrangement portion **21** including the corresponding plate-like positive electrode core body and the corresponding positive electrode active material containing layer **24** that is provided at the core body, and the positive electrode collector tab **22** being integrated with the corresponding positive electrode core body and protruding from the corresponding positive electrode active material arrangement portion **21**. Each positive electrode collector tab **22** includes the corresponding curved portion **22***a* that is positioned between the corresponding positive electrode active material arrangement portion **21** and the positive electrode collector terminal **40** in the Z direction and that has at least a part thereof curved, and the end (the corresponding height-direction extending portion **22***b* and the corresponding flat portion **22***c*) that is positioned at the corresponding curved portion **22***a* on a side opposite to the side of the corresponding positive electrode active material arrangement portion **21**. Each end includes the joint portion that is joined and electrically connected to the positive electrode collector terminal **40**.
(48) Therefore, each positive electrode collector tab **22** includes the corresponding curved portion **22***a* that is positioned between the corresponding positive electrode active material arrangement portion **21** and the positive electrode collector terminal **40** in the Z direction and that has at least a part thereof curved, and each curved portion **22***a* is loose (flexed). Therefore, when the rectangular battery **1** is subjected to shock or vibration, each curved portion **22***a* becomes a shock-absorbing portion, and the positive electrode collector tabs **22** are not excessively pulled. Therefore, it is possible to suppress damage to the positive electrode collector tabs **22**.

(49) The plurality of positive electrode collector tabs **22** that are included in the plurality of positive electrode plates **20** may be disposed so as to overlap each other and may constitute positive-electrode-side tab bundles **41**, and the positive electrode collector terminal **40** may include a plate-like portion **45**. Positive-electrode-side covers (clamping portions) **50** that clamp, along with the edge-portion end surfaces of the plate-like portion **45**, the positive-electrode-side tab bundles **41** may be further provided.
(50) According to the structure above, when the electrode body **14** is inserted into the rectangular outer casing **11**, a part of each positive-electrode-side tab bundle **41** can be clamped by the positive electrode collector terminal **40** and the positive-electrode-side covers **50**, so that it is possible to suppress sideways spreading of the positive-electrode-side tab bundles **41** towards the side of the rectangular outer casing **11**. Therefore, even if the curved portions **22***a* in which the positive-electrode-side tab bundles **41** are loose are provided, it is possible to prevent the positive-electrode-side tab bundles **41** from coming into contact with the side surfaces of the rectangular outer casing **11**. Consequently, even if the curved portions **22***a* in which the positive-electrode-side tab bundles **41** are loose are provided, it is possible to smoothly accommodate the electrode body **14** inside the rectangular outer casing **11**, and to prevent the positive-electrode-side tab bundles **41** from coming into contact with the rectangular outer casing **11** and from becoming damaged. Further, since the positive-electrode-side tab bundles **41** do not come into contact with the side surfaces of the rectangular outer casing **11**, the rectangular battery **1** can be one having high quality.
(51) The positive-electrode-side covers (the clamping portions) **50** and the positive electrode collector terminal **40** may be connected to each other by connecting portions **42**. The positive-electrode-side covers **50**, the positive electrode collector terminal **40**, and the connecting portions **42** may be integrated with each other, and may define recessed portions **43** that accommodate parts of the positive-electrode-side tab bundles **41**.
(52) According to the structure above, the structure that prevents the positive-electrode-side tab bundles **41** from coming into contact with the side surfaces of the rectangular outer casing **11** when inserting the electrode body **14** into the rectangular outer casing **11** can be easily formed at a low cost. Further, when, with the connection portions of the positive-electrode-side covers **50** that are connected with the corresponding connecting portions **42** being fulcra, the positive-electrode-side covers **50** in the XY planes are bent so as to rotate in the directions of arrows C, and thereby a part of each positive-electrode-side tab bundle **41** is clamped by the corresponding positive-electrode-side cover **50** and the positive electrode collector terminal **40** in the X direction, the positive-electrode-side tab bundles **41** can be considerably more easily and reliably clamped by the positive-electrode-side covers **50** and the positive electrode collector terminal **40**.
(53) Further, each positive electrode collector tab **22** may include a flat portion **22***c* that extends along the upper surface **46** of the plate-like portion **45** of the positive electrode collector terminal **40** on the side of the cover plate **12** in the Z direction. When seen from the Z direction, overlapping portions **48** where all of the flat portions **22***c* in all of the positive electrode collector tabs **22** included in the positive-electrode-side tab bundles **41** overlap each other may exist. The joint portion of each positive electrode collector tab **22** that is joined to the positive electrode collector terminal **40** may be included in the overlapping portions **48**, and the length of each overlapping portion **48** in the protruding direction of the positive electrode collector tabs **22** may be greater than or equal to 3 mm.
(54) According to the structure above, since the length of each overlapping portion **48** in the protruding direction of the positive electrode collector tabs **22** is greater than or equal to 3 mm, all of the positive electrode collector tabs **22** included in the positive-electrode-side tab bundles **41** can be reliably joined to the plate-like portion **45**.
(55) The present disclosure is not limited to the embodiment and the modifications thereof above, and various improvements and changes are possible within the scope of the matters described in the claims of the present application and within the scope of equivalents thereof.

(56) For example, in the embodiment above, (i) regarding the positive electrode side, the structure in which each positive electrode collector tab **22** is positioned between the corresponding positive electrode active material arrangement portion **21** and the positive electrode collector terminal **40** in the Z direction, and includes the corresponding curved portion **22***a* that has at least a part thereof curved is described. In addition, (ii) the structure in which a part of each positive-electrode-side tab bundle **41** is clamped by the corresponding positive-electrode-side cover (the clamping portion) **50** and the positive electrode collector terminal **40** is described. Further, (iii) the structure in which the positive-electrode-side covers (the clamping portions) **50** and the positive electrode collector terminal **40** are integrated with each other is described. Further, there is described (iv) the structure in which each positive electrode collector tab **22** includes the corresponding flat portion **22***c* that extends along the upper surface **46** of the plate-like portion **45** of the positive electrode collector terminal **40** on the side of the cover plate **12** in the Z direction; in which when seen from the Z direction, the overlapping portions **48** where all of the entire flat portions **22***c* in all of the positive electrode collector tabs **22** included in the positive-electrode-side tab bundles **41** overlap each other exist; in which the joint portion of each positive electrode collector tab **22** that is joined to the positive electrode collector terminal **40** is included in the overlapping portions **48**; and in which the length of each overlapping portion **48** in the protruding direction of the positive electrode collector tabs **22** is greater than or equal to 3 mm.
(57) However, the rectangular battery need not include, of the four structures (i), (ii), (iii), and (iv), three structures (ii), (iii), and (iv) excluding the structure (i) described first in which each positive electrode collector tab includes the corresponding curved portion. One or more of the four structures (i), (ii), (iii), and (iv) may be provided on the negative electrode side. As long as the negative electrode side includes one of more of the four structures, even on the negative electrode side, it is possible to acquire one or more of the operational effects corresponding to one or more operational effects that are provided by the corresponding one or more of the structures on the positive electrode side. Of the four structures above, the structure (i) described first in which each collector tab includes the corresponding curved portion may be provided only on the positive electrode side, only on the negative electrode side, or on both the positive electrode side and the negative electrode side.
(58) The case in which with the connection portions of the positive-electrode-side covers (the clamping portions) **50** that are connected with the corresponding connecting portions **42** being fulcra, the positive-electrode-side covers **50** are bent in XY planes so as to rotate in the directions of arrows C, and thereby a part of each positive-electrode-side tab bundle **41** is clamped by the corresponding positive-electrode-side cover **50** and the corresponding edge portion of the positive electrode collector terminal **40** in the X direction is described. However, as shown in FIG. **11**, that is, a perspective view of an integral structure of clamping portions **150** and a collector terminal **140** in a modification, it is possible to use a structure in which on at least one of the positive electrode side and the negative electrode side, with connection portions of the clamping portions **150** that are connected with connecting portions **142** being fulcra, the clamping portions **150** are bent in YZ planes so as to rotate in the directions of arrows E, and thereby a part of each tab bundle (not shown) is clamped by the corresponding clamping portion **150** and a corresponding edge portion **141** of the collector terminal **140** in the X direction.
(59) The case in which the positive-electrode-side covers (the clamping portions) **50** are connected to the positive electrode collector terminal **40** by the connecting portions **42** and the positive-electrode-side covers **50** and the positive electrode collector terminal **40** are integrated with each other is described. However, on at least one of the positive electrode side and the negative electrode side, the clamping portions need not be integrated with the collector terminal and may be a single member. In this case, the clamping portions may be made of a metal or a material other than a metal, such as a resin. Alternatively, the clamping portions need not be integrated with the collector terminal and need not be a single member, and may be integrated with an insulator (a member

corresponding to the insulating cover **60** (refer to FIG. **3**)) that is fitted into the rectangular outer casing. In this case, when the clamping portions are made of the same material as the insulator (the insulating cover), the clamping portions may be integrated with the insulator (the insulating cover). Alternatively, when the clamping portions are made of a metal, the clamping portions may be integrated with the insulator made of, for example, a resin by being joined (fixed) to the insulator by using an adhesive or performing welding.
(60) The case in which the plurality of positive electrode collector tabs **22** are separated in two and are disposed so as to overlap each other and form two bundles is described. However, on at least one of the positive electrode side and the negative electrode side, the collector tabs may be formed in one bundle, or may be disposed so as to be separated in three or more, and form three or more bundles. Here, when the collector tabs are formed in one bundle, as shown in FIG. **12**, the collector tabs can be joined to a collector terminal. In FIG. **12**, reference numeral **222** denotes the collector tabs, reference numeral **240** denotes the collector terminal, reference numeral **241** denotes a tab bundle, and reference numeral **250** denotes a clamping portion. According to this modification, since all of the plurality of collector tabs **222** only need to be collector foils on one side in the X direction, it is possible to efficiently join the collector tabs **222** to the collector terminal **240** in a short time.
REFERENCE SIGNS LIST
(61) **11** rectangular outer casing **12** cover plate **14** electrode body **18** positive-electrode-side external terminal **19** negative-electrode-side external terminal **20** positive electrode plate **21** positive electrode active material arrangement portion **22** positive electrode collector tab **22***a* curved portion **22***b* height-direction extending portion **22***c* flat portion **24** positive electrode active material containing layer **30** negative electrode plate **40** positive electrode collector terminal **41** positive-electrode-side tab bundle **42**, **142** connecting portion **43** recessed portion **45** plate-like portion of positive electrode collector terminal **46** upper surface of plate-like portion **48** overlapping portion **50** positive-electrode-side cover **60** insulating cover **140**, **240** collector terminal **150**, **250** clamping portion **222** collector tab **241** tab bundle X direction thickness direction of positive electrode plate (stacking direction) Z direction height direction of rectangular outer casing

Claims

1. A rectangular secondary battery comprising: an electrode body that includes a plurality of stacked electrode plates; a rectangular outer casing that has an opening and that accommodates the electrode body; a cover plate that seals the opening and that is provided with an external terminal on a side opposite to a side of the electrode body; and a collector terminal that is electrically connected to the external terminal and that is disposed inside the rectangular outer casing, wherein each electrode plate includes an active material arrangement portion and a collector tab, the active material arrangement portion including a plate-like core body and an active material containing layer that is provided at the core body, the collector tab being integrated with the core body and protruding from the active material arrangement portion, wherein each collector tab includes a curved portion and an end, the curved portion being positioned between the active material arrangement portion corresponding thereto and the collector terminal in a height direction of the rectangular outer casing and having at least a part thereof curved, the end being positioned at the curved portion on a side opposite to a side of the active material arrangement portion, wherein the end of the collector tab includes a portion thereof disposed directly on a surface of the collector terminal facing away from the active material arrangement portion of each electrode plate, wherein each end includes a joint portion that is joined and electrically connected to the collector terminal, wherein the collector tabs that are included in the plurality of electrode plates are disposed so as to overlap each other and constitute a tab bundle, wherein the tab bundle includes a proximal end

thereof directly connected to the electrode body, and a distal end thereof directly connected to the collector terminal and including the joint portion, and the proximal end and the distal end of the tab bundle overlap each other in the height direction, wherein the collector terminal includes a plate-like portion having a major surface directly adjacent to the joint portion of each end of the collector tab, and an edge-portion end surface extending in a thickness direction of the plate-like portion and apart from the joint portion of each end of the collector tab, wherein the rectangular secondary battery further comprises a clamping portion having a surface thereof opposite the edge-portion end surface of the plate-like portion and extending in the thickness direction of the plate-like portion, so as to clamp a portion of the tab bundle apart from the joint portion, wherein said portion of the tab bundle is directly sandwiched, in a direction perpendicular to the thickness direction of the plate-like portion, between the surface of the clamping portion and the edge-portion end surface of the plate-like portion.

2. The rectangular secondary battery according to claim 1, wherein the rectangular secondary battery further comprises: a connecting portion configured to pivotably connect the clamping portion to the collector terminal, wherein the clamping portion, the collector terminal, and the connecting portion are integrated with each other and define a recessed portion that accommodates a part of the tab bundle.

3. The rectangular secondary battery according to claim 1, comprising: an insulator that is disposed between an inner surface portion of the rectangular outer casing and the clamping portion, the inner surface portion being positioned on a side opposite to a side of the edge-portion end surface with respect to the clamping portion, wherein the clamping portion is integrated with the insulator, or is fixed to the insulator.

4. The rectangular secondary battery according to claim 1, wherein each collector tab includes a flat portion that extends along an upper surface of the plate-like portion on a side of the cover plate in the height direction, wherein, when seen from the height direction, an overlapping portion where all of the flat portions in all of the collector tabs included in the tab bundle overlap each other exists, wherein the joint portion is included in the overlapping portion, and wherein a length of the overlapping portion in a protruding direction of the collector tabs is greater than or equal to 3 mm.

5. The rectangular secondary battery according to claim 1, further comprising: an insulating cover having a substantially U-shaped cross section press-fitted to inner surfaces of a pair of side plates of the rectangular outer casing, wherein the insulating cover is disposed between an inner surface portion of the rectangular outer casing and the clamping portion, the inner surface portion facing the edge-portion end surface of the plate-like portion via the clamping portion, so as to force the clamping portion toward the edge-portion end surface of the plate-like portion.